

Architectural Roles of Protein Condensates across DNA, Cytoskeletal, and Membrane Systems

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Protein condensates can do more than concentrate biomolecules: they can organize and physically couple larger structures. In this talk, I will discuss three examples from reconstituted biophysical systems. First, tau binds and compacts DNA into dynamic co-condensates that capture and organize microtubules, suggesting a possible link to the centromeric localization of tau observed during mitosis. Second, tau–tubulin co-condensates promote microtubule nucleation and dynamically weave growing networks through bundling, crosslinking, and remodeling. Finally, using freestanding lipid bilayers reconstituted with neuronal SNAREs, we find that synapsin condensates cluster VAMP2-bearing vesicles while promoting their association with target membranes; condensate dispersal then releases the vesicles for SNARE-mediated fusion. Together, these results illustrate how condensates can act as dynamic mesoscale architects across DNA, cytoskeletal, and membrane systems.

[References]

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